

A DIMENSION-FREE COMPARISON BETWEEN EXPECTATION THRESHOLDS AND FRACTIONAL EXPECTATION THRESHOLDS

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ABSTRACT. We prove a dimension-free comparison between the expectation threshold $q(\mathcal{F})$ and the fractional expectation threshold $q_f(\mathcal{F})$ for any nontrivial increasing family $\mathcal{F}$ on a finite ground set. Specifically, we show that there is a universal constant $K > 0$ such that

$$q_f(\mathcal{F}) \leq K q(\mathcal{F}) \max\{1, \log \log(1/q(\mathcal{F}))\}.$$

Combining this comparison with a recent result of Li [arXiv:2609.08967] on the fractional version of Talagrand's discrete Convexity Conjecture, we obtain a dimension-independent bound toward the conjecture.

1. INTRODUCTION

In this note, we show a dimension-free comparison between the expectation threshold and the fractional expectation threshold (Theorem 1.2 below). The comparison follows rather quickly from [13, Theorem 1.3], a quantitative strengthening of [11, Theorem 1.5].

We first briefly recall the necessary definitions. (For a more detailed introduction to the background, see, e.g., [3, 8].) All logarithms are natural, unless a base is explicitly indicated.

We use V for a finite set, and write 2^V for its power set. A family $\mathcal{F} \subseteq 2^V$ is called *increasing* if $A \in \mathcal{F}$ and $B \supseteq A$ imply $B \in \mathcal{F}$. For $\mathcal{G} \subseteq 2^V$ we use $\langle \mathcal{G} \rangle$ for the increasing family generated by $\mathcal{G}$, namely $\{B \subseteq V : \exists A \in \mathcal{G}, B \supseteq A\}$. Throughout, we assume that $\mathcal{F} \subseteq 2^V$ is increasing and nontrivial, i.e., $\mathcal{F} \not\subseteq \{\emptyset, 2^V\}$.

We call $\mathcal{F}$ *p-small* if there exists a $\mathcal{G} \subseteq 2^V$ such that

$$\langle \mathcal{G} \rangle \supseteq \mathcal{F} \text{ and } \sum_{S \in \mathcal{G}} p^{|S|} \leq 1/2,$$

and the *expectation threshold* of $\mathcal{F}$ is $q(\mathcal{F}) = \max\{p : \mathcal{F} \text{ is } p\text{-small}\}$. Say $\mathcal{F}$ is *weakly p-small* if there is a function $\lambda : 2^V \rightarrow \mathbb{R}^+ (= [0, \infty))$ such that

$$(1) \quad \sum_{S \subseteq I} \lambda_S \geq 1 \quad \forall I \in \mathcal{F} \quad \text{and} \quad \sum_S \lambda_S p^{|S|} \leq 1/2.$$

We then define the *fractional expectation threshold* of $\mathcal{F}$ by $q_f(\mathcal{F}) = \max\{p : \mathcal{F} \text{ is weakly } p\text{-small}\}$.

It follows immediately from the definition that $q(\mathcal{F}) \leq q_f(\mathcal{F})$ (by taking the function λ to be the indicator function of $\mathcal{G}$). The following conjecture of Talagrand [15, Conjecture 6.3] asserts a reverse inequality up to a universal constant.

Conjecture 1.1. *There is a universal $L > 0$ such that for every finite set V and increasing family $\mathcal{F} \subseteq 2^V$,*

$$q_f(\mathcal{F}) \leq L q(\mathcal{F}).$$

Equivalently,

$$(2) \quad \text{weakly } p\text{-small implies } (p/L)\text{-small.}$$

Our main goal is to establish the following dimension-free comparison between q and q_f .

Theorem 1.2. *There is a universal $K > 0$ such that for every finite set V and increasing family $\mathcal{F} \subseteq 2^V$,*

$$q_f(\mathcal{F}) \leq Kq(\mathcal{F}) \cdot \max\{1, \log \log(1/q(\mathcal{F}))\}.$$

We will in fact prove the following equivalent statement (up to changing the universal constant K):

- (3) for every $p \in (0, 1)$, if $\mathcal{F}$ is weakly p -small, then it is $p/(K \cdot \max\{1, \log \log(1/p)\})$ -small.

Proof of the equivalence. Write q, q_f for $q(\mathcal{F}), q_f(\mathcal{F})$. That (3) implies Theorem 1.2 follows immediately from $q \leq q_f$ and the fact that $f(t) = \max\{1, \log \log(1/t)\}$ is non-increasing.

To see the converse: fix the universal constant $K \geq 1$ in Theorem 1.2, and suppose that $\mathcal{F}$ is weakly p -small. Since Conjecture 1.1 holds when q is bounded away from zero (after adjusting the constant L), we may assume that q is sufficiently small, say, $q \leq \min\{1/100, 1/K^3\}$. Thus Theorem 1.2 gives that $p \leq q_f \leq Kq \log \log(1/q)$, and hence

$$\log(1/p) \geq \log(1/q) - \log K - \log \log \log(1/q) \geq \frac{1}{2} \log(1/q).$$

Therefore,

$$p \leq 2Kq \log \log(1/p) \leq 2Kq \max\{1, \log \log(1/p)\}.$$

□

To place Theorem 1.2 in context, we first review some related background. Conjecture 1.1 has motivated a number of partial results. In particular, the conjectured constant-factor comparison has been established under various additional assumptions on the family $\mathcal{F}$ or the function λ [3, 8, 5, 4, 13, 7, 6].

In the unrestricted setting, the best previously known general bound is obtained by combining the results of Pham [13, Theorem 1.2] and Fischer and Person [5, Theorem 10]:

$$(4) \quad q_f(\mathcal{F}) \leq Cq(\mathcal{F}) \cdot \max\{1, \log \log |V|\}^1.$$

We remark that Theorem 1.2 strengthens (4), up to a change in the universal constant. To see this, note that (4) is equivalent to

$$(5) \quad \text{for every } p \in (0, 1), \text{ weakly } p\text{-small implies } p/(C \cdot \max\{1, \log \log |V|\})\text{-small.}$$

If $p \leq 1/(2|V|)$, taking $\mathcal{G}$ to be the family of all singleton subsets of V shows that every nontrivial increasing family is p -small. Therefore, even the conclusion of (2) automatically holds in this case. On the other hand, if $p > 1/(2|V|)$, then

$$\max\{1, \log \log(1/p)\} = O(\max\{1, \log \log |V|\}).$$

Thus, (3) implies (5).

The primary motivation for seeking a dimension-free comparison was the discrete version of Talagrand's Convexity Conjecture (see, e.g., [15, 1]) – see Remark 3.3. In Section 3, we derive a weaker version of the discrete Convexity Conjecture (Corollary 3.5), while retaining dimension independence.

Our derivation is inspired by the approach of [4]. We replace their Lemma 1.3 (which is essentially [2, Lemma 3.4]) with [13, Theorem 1.3]. A similar argument appears in [13, Section 3]. We next restate [13, Theorem 1.3] in the form that we will use; this requires a little additional preparation.

Let $\mathcal{F} \subseteq 2^V$ be given. For each $I \in \mathcal{F}$, let μ_I be a probability measure on I ; that is, $\mu_I(v) \geq 0$ for every $v \in I$ and $\sum_{v \in I} \mu_I(v) = 1$. For $A \subseteq I$, write $\mu_I(A) := \sum_{v \in A} \mu_I(v)$. We say that $Y \subseteq V$ is c -bad ($c > 0$) if

$$\sup_{I \in \mathcal{F}} \mu_I(Y \cap I) < c.$$

¹Here we assume $|V| \geq 2$ to ensure that $\log \log |V|$ is meaningful.

For $r \in [0, 1]$, let V_r denote a random subset of V obtained by including each element independently with probability r . With this notation, [13, Theorem 1.3] can be stated as follows:

Theorem 1.3. *Let $q \in (0, 1)$, and suppose $\mathcal{F}$ is not q -small. If $s \geq 1$ is an integer and $16sq \leq 1$, then*

$$\mathbb{P}(V_{16sq} \text{ is } (1 - 2^{-s})\text{-bad}) \leq 2/3.$$

2. PROOF OF THEOREM 1.2

We prove (3). Our proof initially follows the argument of [4]. Fix $p \in (0, 1)$, and set

$$s = \lceil \log_2 \log_2(32/p) \rceil + 1, \quad q = p/(512s), \quad \rho = 16sq.$$

For convenience, write L for 2^s .

Suppose that $\mathcal{F}$ is weakly p -small, and let λ satisfy (1). We may (and will) assume $\lambda_\emptyset = 0$; indeed, if $\lambda_\emptyset > 0$, then define λ' by

$$\lambda'_\emptyset = 0 \text{ and } \lambda'_S = \lambda_S/(1 - \lambda_\emptyset) \text{ if } S \neq \emptyset.$$

Then λ' also satisfies (1) (with $1/2$ improved to $(1 - 2\lambda_\emptyset)/(2(1 - \lambda_\emptyset))$).

For each $I \in \mathcal{F}$, set

$$(6) \quad \nu_I := \sum_{S \subseteq I} |S| \lambda_S.$$

Since $\lambda_\emptyset = 0$ and $\sum_{S \subseteq I} \lambda_S \geq 1$, we have $\nu_I \geq 1$. Define $\mu_I : I \rightarrow \mathbb{R}^+$ by

$$\mu_I(v) = \nu_I^{-1} \sum_{v \in S \subseteq I} \lambda_S \quad \forall v \in I$$

and notice that $\mu_I(I) = 1$.

Suppose, for a contradiction, that $\mathcal{F}$ is not q -small. Then by Theorem 1.3 and the definition of ρ , the probability that V_ρ is $(1 - 2^{-s})$ -bad (with respect to the μ_I 's) is at most $2/3$. We will obtain a contradiction by proving the following lower bound.

Claim 2.1. With probability at least $3/4$, V_ρ is $(1 - 2^{-s})$ -bad with respect to the μ_I 's.

Proof. For any $Y \subseteq V$ and $I \in \mathcal{F}$,

$$\mu_I(Y \cap I) = \nu_I^{-1} \sum_{v \in Y \cap I} \sum_{v \in S \subseteq I} \lambda_S = \nu_I^{-1} \sum_{S \subseteq I} |S \cap Y| \lambda_S = 1 - \nu_I^{-1} \sum_{S \subseteq I} |S \setminus Y| \lambda_S.$$

Here we depart from the argument of [4]. For $Y \subseteq V$, define²

$$\mathcal{B}(Y) = \{S \subseteq V : |S \setminus Y| < 2|S|/L\}.$$

Since $S \notin \mathcal{B}(Y)$ implies $L|S \setminus Y|/2 \geq |S|$,

$$\begin{aligned} \nu_I^{-1} \sum_{S \subseteq I} |S \setminus Y| \lambda_S &\geq \nu_I^{-1} \sum_{S \subseteq I, S \notin \mathcal{B}(Y)} |S \setminus Y| \lambda_S \\ &\geq \frac{2}{L\nu_I} \sum_{S \subseteq I, S \notin \mathcal{B}(Y)} |S| \lambda_S \\ &\stackrel{(6)}{=} \frac{2}{L} - \frac{2}{L\nu_I} \sum_{S \subseteq I, S \in \mathcal{B}(Y)} |S| \lambda_S. \end{aligned}$$

²For readers familiar with the proof of [4]: $\mathcal{B}(Y)$ plays a role analogous to that of the family $\{S : S \subseteq Y\}$ in [4]. The assumption on λ there allows us to restrict attention to sets S with $|S| \leq r$. For such sets, $S \not\subseteq Y$ implies $|S \setminus Y| \geq |S|/r$, a crucial point in that proof. Here, the analogous bound $|S \setminus Y| \geq 2|S|/L$ holds whenever $S \notin \mathcal{B}(Y)$.

When $Y = V_\rho$, we may bound the last sum (for any I) by $\sum_{S \in \mathcal{B}(V_\rho)} |S| \lambda_S$, so it remains to prove the following bound.

Claim 2.2. Write $X = V_\rho$. With probability at least $3/4$, $\sum_{S \in \mathcal{B}(X)} |S| \lambda_S < 1/2$.

Proof. Note that $S \in \mathcal{B}(Y)$ iff $|S \cap Y| > (1 - 2/L)|S|$. Therefore, for $S \in \binom{V}{k}$,

$$\mathbb{P}(S \in \mathcal{B}(X)) \leq \binom{k}{k - \lfloor 2k/L \rfloor} \rho^{k - \lfloor 2k/L \rfloor} \leq 2^k \rho^{k(1 - 2/L)} \leq (4\rho)^k$$

where the last inequality holds because, by our choice of parameters, $\rho^{-2/L} \leq 2$. Since $k \leq 2^k$,

$$k \cdot \mathbb{P}(S \in \mathcal{B}(X)) \leq (8\rho)^k.$$

Thus

$$\mathbb{E} \left[\sum_{S \in \mathcal{B}(X)} |S| \lambda_S \right] = \sum_{S \neq \emptyset} \mathbb{P}(S \in \mathcal{B}(X)) |S| \lambda_S \leq \sum_{S \neq \emptyset} \lambda_S (8\rho)^{|S|} = \sum_{S \neq \emptyset} \lambda_S (p/4)^{|S|} \leq \frac{1}{4} \sum_S \lambda_S p^{|S|} \leq \frac{1}{8}.$$

Now apply Markov's inequality. ■

By Claim 2.2 (and since $\nu_I \geq 1$ for any $I \in \mathcal{F}$), with probability at least $3/4$,

$$\sup_{I \in \mathcal{F}} \mu_I(V_\rho \cap I) < 1 - \frac{1}{L},$$

i.e., V_ρ is $(1 - 1/L)$ -bad. □

Claim 2.1 gives the desired contradiction, so $\mathcal{F}$ is q -small. Finally, since $s = O(\max\{1, \log \log(1/p)\})$ and $q = p/(512s)$, we obtain the conclusion in (3).

3. DISCUSSION

We now discuss the connection with the discrete version of Talagrand's Convexity Conjecture. For $p \in [0, 1]$, let μ_p denote the distribution of V_p on 2^V ; that is, for every $A \subseteq V$, $\mu_p(\{A\}) = p^{|A|}(1-p)^{|V \setminus A|}$, and $\mu_p(\mathcal{A}) = \mathbb{P}(V_p \in \mathcal{A}) = \sum_{A \in \mathcal{A}} \mu_p(\{A\})$ for every $\mathcal{A} \subseteq 2^V$.

For a family $\mathcal{A} \subseteq 2^V$ and a positive integer k , set

$$\mathcal{A}_{(k)} := \{S \subseteq V : S \subseteq A_1 \cup \dots \cup A_k \text{ for some } A_1, \dots, A_k \in \mathcal{A}\},$$

and $\mathcal{A}^{(k)} := 2^V \setminus \mathcal{A}_{(k)}$. Thus $\mathcal{A}^{(k)}$ is the increasing family of sets not contained in any union of k members of $\mathcal{A}$.

Conjecture 3.1 (Talagrand [14, 15]). *There exist a positive integer k and a universal constant $L \geq 1$ such that, for every $p \in (0, 1)$, finite set V , and family $\mathcal{A} \subseteq 2^V$, if $\mu_p(\mathcal{A}) \geq 1 - 1/(2k)$ then $\mathcal{A}^{(k)}$ is (p/L) -small.*

Hua, Song, and Tudose [9] resolved the original Gaussian version of Talagrand's Convexity Conjecture. Using an argument of Talagrand, they deduced the following partial result bound toward Conjecture 3.1; see [9, Corollary 1.4 and Appendix A].

Theorem 3.2. *There exist $\epsilon > 0$ and $L \geq 1$ so that for any finite set V , $p \in (0, 1)$, and any family $\mathcal{A} \subseteq 2^V$, if $\mu_p(\mathcal{A}) \geq 1 - \epsilon$ then $\mathcal{A}^{(3)}$ is p^L -small.*

For comparison, a result of Pham and the author [12, Theorem 1.1] implies that, for any $\mathcal{A} \subseteq 2^V$ with $\mu_p(\mathcal{A}) \geq 1/2$, the family $\mathcal{A}^{(1)}$ is $(p/(C \log \ell(\mathcal{A})))$ -small for a universal constant C , where $\ell(\mathcal{A})$ is the maximum of 2 and the size of a largest minimal element of $\mathcal{A}^{(1)}$. Since $\mathcal{A}^{(3)} \subseteq \mathcal{A}^{(1)}$, this also gives the same

smallness bound for $\mathcal{A}^{(3)}$. The two bounds from [9] and [12] are not uniformly comparable; we refer the readers to the discussion following [9, Corollary 1.4] for further discussion.

Remark 3.3. A key requirement of Conjecture 3.1 is that the smallness parameter depend only on the density parameter p , independently of the size of the ground set. The importance of dimension independence was emphasized by Talagrand in [15, Section 1].

Recently, Li [10, Corollary 3.1] proved a fractional version of Conjecture 3.1. His result is formulated in terms of spread probability measures. By the duality relation between spread measures and being weakly-small [15, Proposition 6.7], it yields the following statement in our terminology:

Theorem 3.4. *For every finite set V , $p \in (0, 1)$, and family $\mathcal{A} \subseteq 2^V$ satisfying $\mu_p(\mathcal{A}) > 1/2$, the family $\mathcal{A}^{(2)}$ is weakly $(p/2)$ -small.*

Combining Theorem 1.2 with Theorem 3.4, we obtain the following dimension-free bound.

Corollary 3.5. *There is a universal $L > 0$ such that, for every finite set V , $p \in (0, 1)$, and $\mathcal{A} \subseteq 2^V$ satisfying $\mu_p(\mathcal{A}) > 1/2$, the family $\mathcal{A}^{(2)}$ is $p/(L \max\{1, \log \log(1/p)\})$ -small.³*

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³By extending Li's argument to $\mathcal{A}^{(2m)}$ for integers $m \geq 1$ and combining this with Theorem 1.2, one also obtains the conclusion that $\mathcal{A}^{(2m)}$ is $\Omega(p/(L \max\{1, \log \log(1/p)\})^{1/m})$ -small, where the implicit constant is universal.

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